

Stereo Camera-Based Environment Perception for Automated Vehicles

Stereokamera-Basierte Umfeldwahrnehmung für automatisierte
Fahrzeuge



Scientific work for obtaining the academic degree
Master of Science (M.Sc.)
at the Department Mobility Systems Engineering
of the TUM School of Engineering and Design
at the Technical University of Munich

Supervised by	Prof. Dr.-Ing. Markus Lienkamp David Brecht, M.Sc. Chair of Automotive Technology
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Submitted on	July 01, 2025

Project description

Titel der Arbeit

Hier steht ein zum Thema hinführender Text.

In einer konstruktiven (oder theoretischen) **Bachelor/-Semester/-Masterarbeit** sollen (...)

Folgende Punkte sind durch **Herrn/Frau Martin Musterstudent** zu bearbeiten:

- Punkt 1
- Punkt 2

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Die eingereichte Arbeit verbleibt als Prüfungsunterlage im Eigentum des Lehrstuhls.

Announcement date: April 1, 2021

Submission date: October 1, 2021

Prof. Dr.-Ing. Markus Lienkamp

Max Musterbetreuer, M.Sc.

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Garching, den 1. Juli 2025

Adrian Rieker, B.Sc.

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Garching, July 1, 2025

Adrian Rieker, B.Sc.

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Formula Symbols

Formula Symbols	Unit	Description
B_{Fzg}	m	Breite des Fahrzeuges
B_{Rad}	m	Reifenbreite (Breite der Lauffläche)
D_{Rad}	m	Raddurchmesser
H_{Fzg}	m	Höhe des Fahrzeuges
R_{Fzg}	m	Radstand des Fahrzeuges
S_{Fzg}	m	Spur des Fahrzeuges
a_i	-	Gewichtungsfaktoren für die Kostenfunktion J
$\mathbf{c}_{\text{DGL}}$	-	Gleichungsnebenbedingungen, die aus der Diskretisierung der Zustandsdifferentialgleichung resultieren
H	-	Hamilton"-Funktion
E_{pot}	J	Potenzielle Energie
E_m	J	Spezifische Energie

Todo list

1 Introduction

2 State of the Art

2.1 Perception Systems for Autonomous Vehicles

2.2 Algorithms for moving object detection

2.3 State of the Art in the FTM

2.4 Stereo camera technology

2.5 Methods for software containerization

2.6 Requirements for the Thesis

3 Methods and Implementation

- High level explanation on how the work will be structured. - What the perception and criticality pipeline are - Go through each component.

3.1 System Architecture

3.1.1 Perception pipeline

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5 Summary and Conclusion

- High level explanation on how the work will be structured. - What the perception and criticality pipeline are - Go through each component.

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A Chapter Anhang

A.1 Section Anhang

A.1.1 Subsection Anhang

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B Chapter Anhang 2



Figure B.1: Bild im Anhang

Table B.1: Tabelle im Anhang

Spalte 1	Spalte 2
...	...